

HOMEWORK 5

1. Prove that for every $n \geq 5$, the alternating group A_n is the only nontrivial normal subgroup of S_n .
2. Prove the following Useful Counting Result. Let H be a subgroup of a finite group G with $H \neq G$. Suppose that $|G|$ does not divide $[G : H]!$. Then G contains a nontrivial normal subgroup N such that N is a subgroup of H . In particular, G is not simple.
3. Prove that all groups of order $2p^n$ and $4p^n$ are not simple (p is a prime integer).
4. Let G be a nonabelian group of order p^3 (p is a prime integer). Prove that the center $Z(G)$ of G coincides with the commutator subgroup $[G, G]$.
5. Let G be a semidirect product of a cyclic normal subgroup N of order n and an abelian group K . Show that if $|K|$ is relatively prime to $\varphi(n)$ (φ is the Euler function), then G is abelian.
6. Determine the center of the dihedral group D_{2n} .
7. Let K and H be two groups and $f : K \rightarrow \text{Aut}(H)$ a group homomorphism. Prove that for every $g \in \text{Aut}(K)$ the external semidirect products $H \rtimes K$ with respect to f and $f \circ g$ are isomorphic.
8. Let K and H be two groups and $f : K \rightarrow \text{Aut}(H)$ a group homomorphism. Prove that for every $s \in \text{Aut}(H)$ the external semidirect products $H \rtimes K$ with respect to f and $f' : K \rightarrow \text{Aut}(H)$, defined by $f'(k) = s \circ f(k) \circ s^{-1}$ for $k \in K$, are isomorphic.
9. Let G be the external semidirect product of $\mathbb{Z}/n\mathbb{Z}$ and $(\mathbb{Z}/n\mathbb{Z})^\times$ with respect to the identity isomorphism $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Aut}(\mathbb{Z}/n\mathbb{Z}) = (\mathbb{Z}/n\mathbb{Z})^\times$. Prove that G is isomorphic to a subgroup of S_n .
10. Let H be a group and $\varphi \in \text{Aut}(H)$. Prove that there is a group G containing H as a subgroup and an element $g \in G$ such that $\varphi(h) = ghg^{-1}$ for all $h \in H$, i.e., φ extends to an inner automorphism of G .